# Today's Agenda

Connecting descriptive statistics to univariate visualizations

Justin Leinaweaver (Fall 2021)

# Pizza, Politics and Afghanistan

**FREE pizza**

**Friday, September 10th @ noon**

**The plaza outside Stone Chapel**

**Brought to you by the Meador Center for Politics & Citizenship**

# Practice Questions

Calculate the minimum, mean and maximum values of the `Total` score for:

1. The years 2013 and 2021 (filter).
2. Each year in the dataset (group_by).
3. Each region in the dataset (group_by).
4. Both year and region (group_by).

# Practice Questions: Baseline Code

```r
d |>
  summarize(
    Minimum = min(Total),
    Mean = mean(Total),
    Maximum = max(Total)
)
```

| Minimum | Mean | Maximum |
|---|---|---|
| -1 | 58.03077 | 100 |

# Practice Questions: Baseline Code (v2)

```r
d |>
  summarize(
    Minimum = min(Total),
    Mean = round(mean(Total), 1),
    Maximum = max(Total)
)
```

| Minimum | Mean | Maximum |
|---|---|---|
| -1 | 58 | 100 |

# Practice Questions: 2013 and 2021

```r
d |>
   filter(Edition == 2013) |>
   summarize(
     Minimum = min(Total),
     Mean = round(mean(Total), 1),
     Maximum = max(Total)
)
```

| Minimum | Mean | Maximum |
|---|---|---|
| 1 | 59.4 | 100 |

# Practice Questions: By Year

```r
d |>
  group_by(Edition) |>
  summarize(
    Minimum = min(Total),
    Mean = round(mean(Total), 1),
    Maximum = max(Total)
)
```

| Edition | Minimum | Mean | Maximum |
|---|---|---|---|
| 2013 | 1 | 59.4 | 100 |
| 2014 | 1 | 59.4 | 100 |
| 2015 | -1 | 59.0 | 100 |
| 2016 | -1 | 58.7 | 100 |
| 2017 | -1 | 58.0 | 100 |
| 2018 | -1 | 57.6 | 100 |
| 2019 | 0 | 57.3 | 100 |
| 2020 | 0 | 56.7 | 100 |
| 2021 | 1 | 56.2 | 100 |

# Practice Questions: By Region

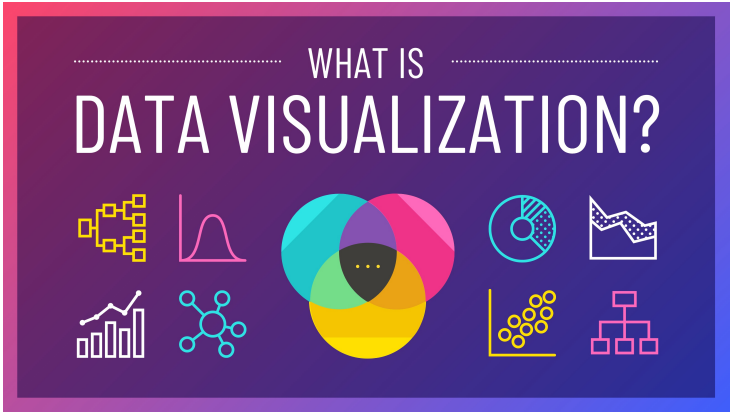
```r
d |>
  group_by(Region) |>
  summarize(
    Minimum = min(Total),
    Mean = round(mean(Total), 1),
    Maximum = max(Total)
)
```

| Region | Minimum | Mean | Maximum |
|---|---|---|---|
| Americas | 11 | 74.3 | 99 |
| Asia | 1 | 58.3 | 99 |
| Eurasia | 2 | 28.5 | 65 |
| Europe | 31 | 87.0 | 100 |
| MENA | -1 | 28.0 | 81 |
| SSA | 2 | 44.1 | 92 |

# Practice Questions: By Region and Year

```r
d |>
  group_by(Region, Edition) |>
  summarize(
    Minimum = min(Total),
    Mean = round(mean(Total), 1),
    Maximum = max(Total)
)
```

| Region | Edition | Minimum | Mean | Maximum |
|---|---|---|---|---|
| Americas | 2013 | 11 | 75.6 | 99 |
| Americas | 2014 | 12 | 75.4 | 99 |
| Americas | 2015 | 14 | 75.3 | 99 |
| Americas | 2016 | 15 | 75.0 | 99 |
| Americas | 2017 | 15 | 74.2 | 99 |
| Americas | 2018 | 14 | 73.8 | 99 |
| Americas | 2019 | 14 | 73.4 | 99 |
| Americas | 2020 | 14 | 72.8 | 98 |
| Americas | 2021 | 13 | 72.7 | 98 |
| Asia | 2013 | 1 | 58.1 | 97 |
| Asia | 2014 | 1 | 58.4 | 97 |
| Asia | 2015 | 1 | 58.4 | 97 |
| Asia | 2016 | 1 | 58.6 | 98 |



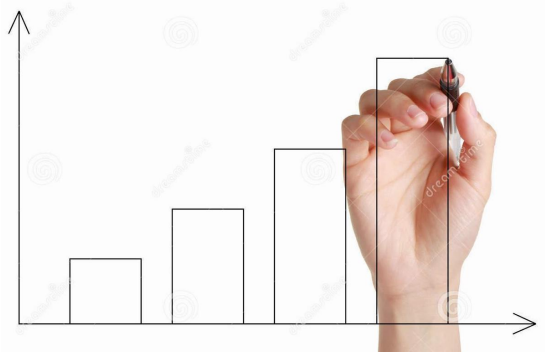

**A method for summarizing data in more compelling, easier to understand and more informative methods than descriptive statistics.**

# Principle 3. Variable Type Determines Tool

**Univariate Visualizations**

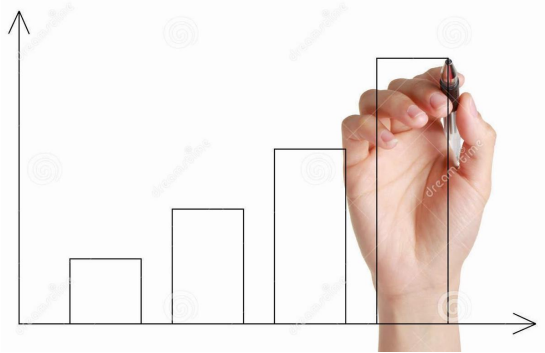
- If a **categorical** variable, make a **bar plot**
- If a **numeric** variable, make a **box plot** or **histogram**
- If a **numeric variable across time**, make a **line plot**

# Why draw visualizations by hand?



1. Principle 3 in action,

# Why draw visualizations by hand?



1. Principle 3 in action,
2. Learn to think like a programmer

# Why draw visualizations by hand?

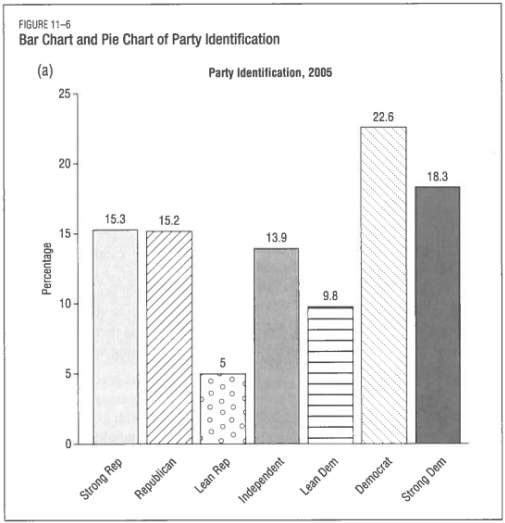
| Variable | N | Mean | StdDev | Min | Max |
|---|---|---|---|---|---|
| CL | 1885 | 35.1 | 17.1 | 1 | 60 |
| CL rating | 1885 | 3.4 | 1.9 | 1 | 7 |
| PR | 1885 | 23.0 | 13.4 | -3 | 40 |
| PR rating | 1885 | 3.6 | 2.2 | 1 | 7 |
| Total | 1885 | 58.0 | 30.2 | -1 | 100 |

1. Principle 3 in action,
2. Learn to think like a programmer
3. Learn to "see" descriptive stats tables

# Categorical Variable: Make a Bar Plot

## Bar Plots

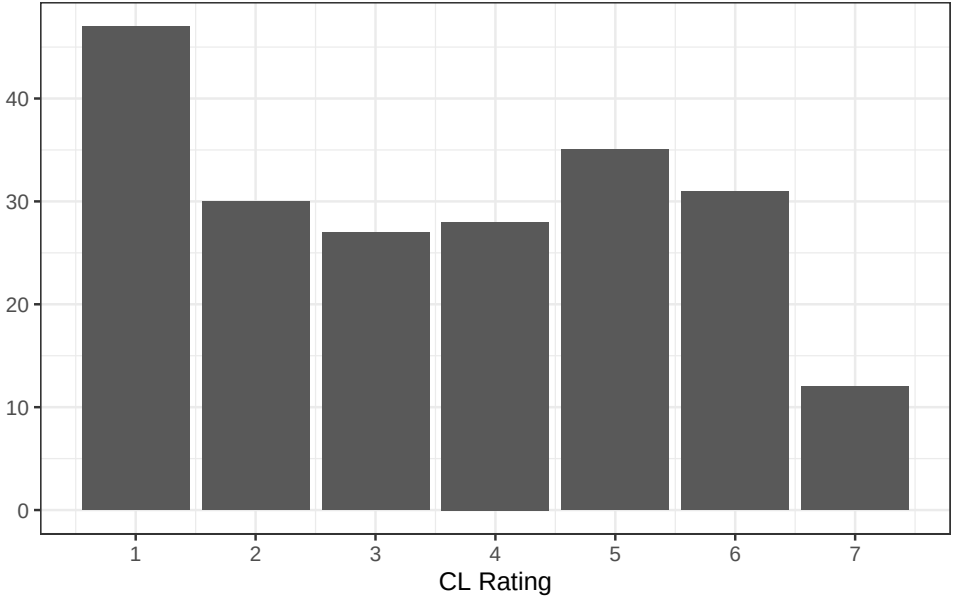
1. Counts, or
2. Proportions



# Categorical Variable: Make a Bar Plot

**Draw a bar plot of `CL Rating` counts in 2021 by hand.**

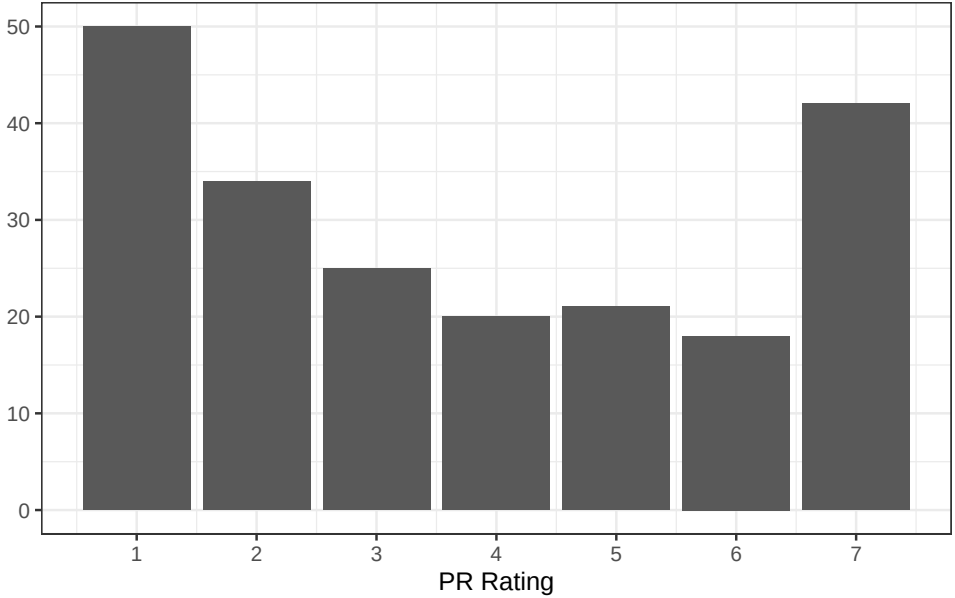
```
table(d$"CL rating")
```


40
30
20
10
0
1
2
3
4
5
6
7
CL Rating

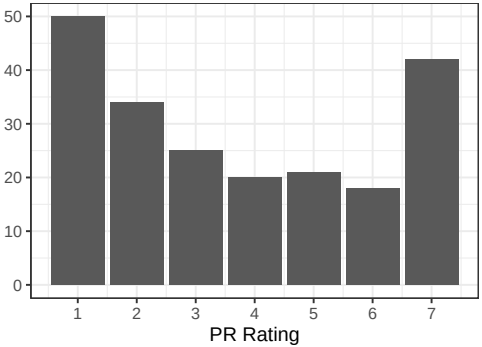
# Categorical Variable: Make a Bar Plot

**Draw a bar plot of "PR Rating" counts in 2021 by hand.**

```
table(d$"PR rating")
```


50
40
30
20
10
0
1
2
3
4
5
6
7
PR Rating

# Bar Plots vs Descriptive Stats Table



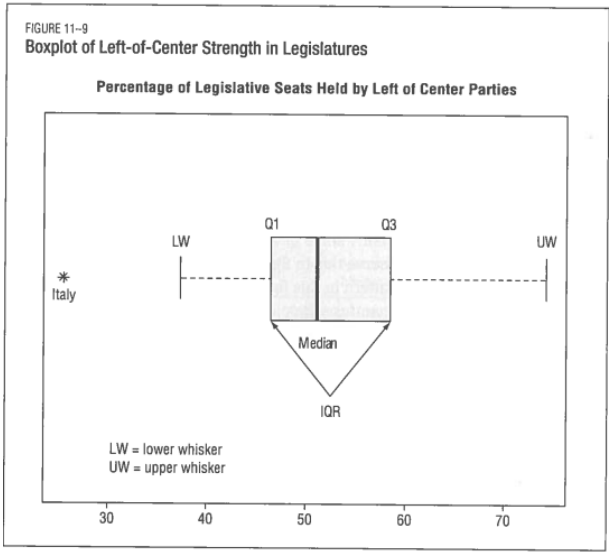

```r
table(d$`PR rating`)
```

| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|
| 501 | 277 | 254 | 176 | 167 | 224 | 286 |

# Numerical Variable: Make a Box Plot

## Box Plots

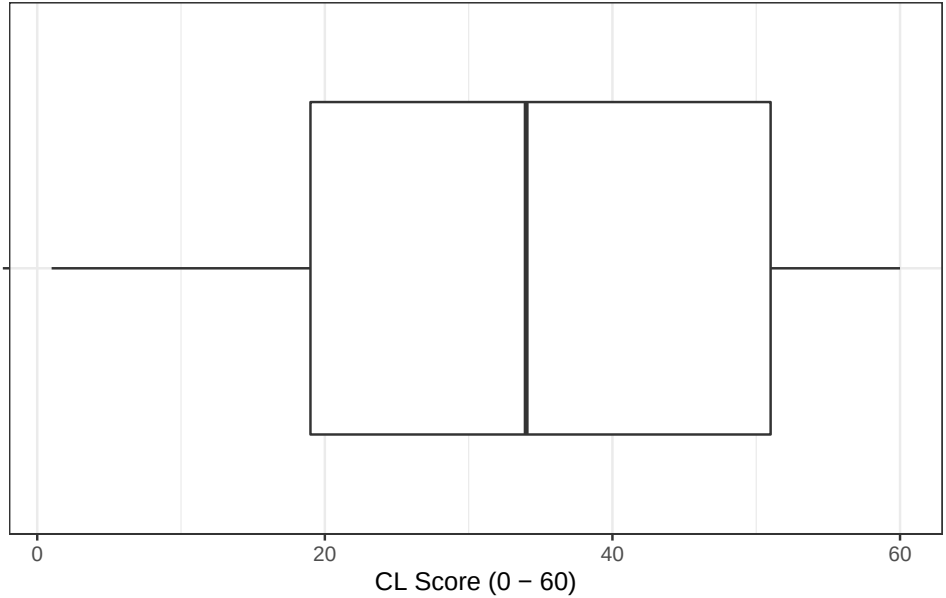
- Minimum
- 25th percentile
- Median
- 75th percentile
- Maximum
- IQR x 1.5



# Numerical Variable: Make a Box Plot

**Draw a box plot of `CL` score in 2021 by hand.**

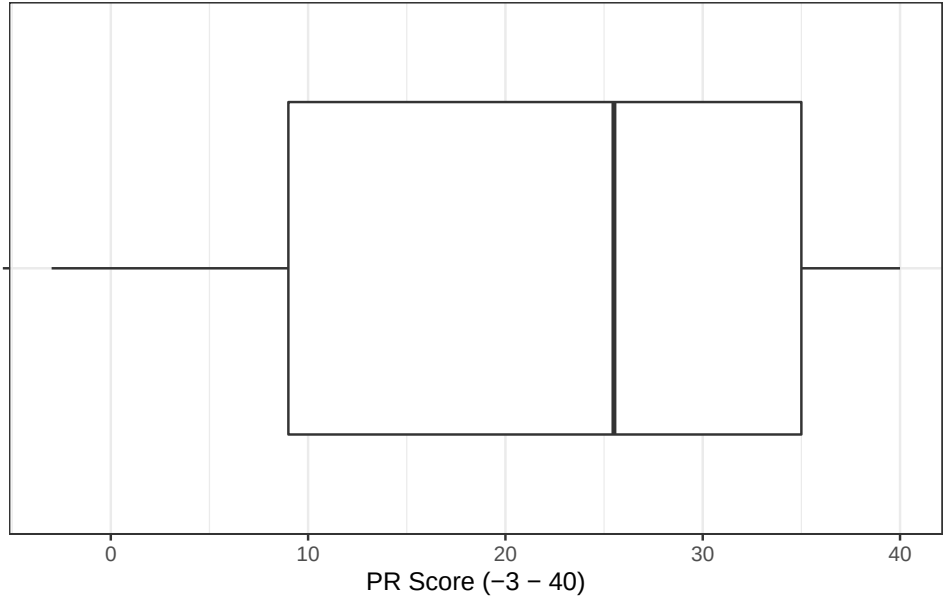
**Note:** Johnson (2012), p388-389 has the steps...


0
20
40
60
CL Score (0 − 60)

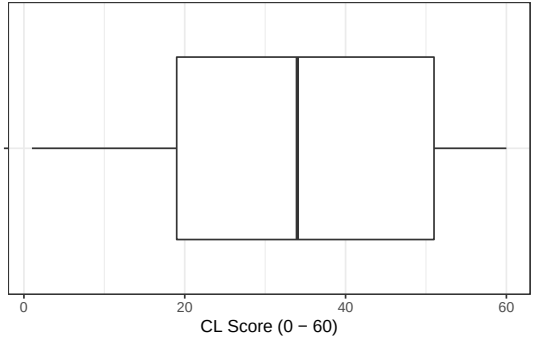
# Numerical Variable: Make a Box Plot

**Draw a box plot of `PR` score in 2021 by hand.**

**Note:** Johnson (2012), p388-389 has the steps...


0
10
20
30
40
PR Score (−3 − 40)

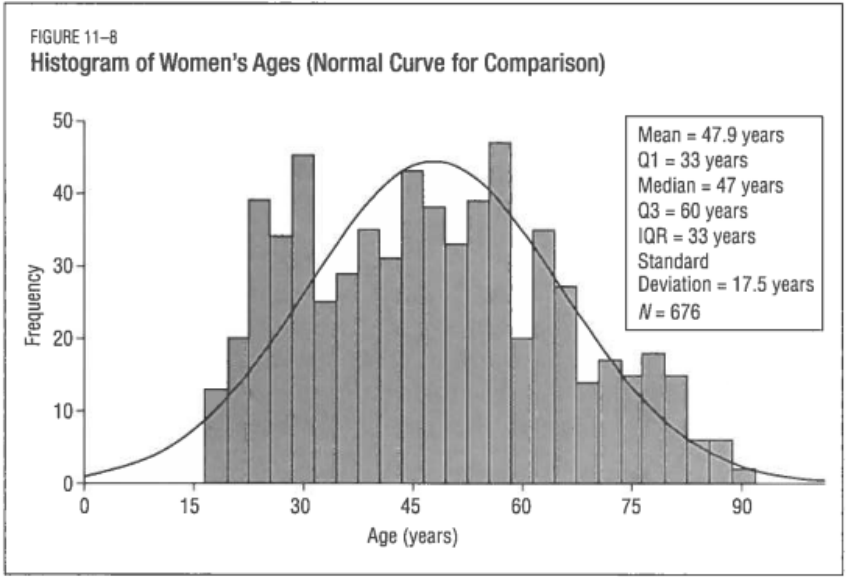
# Box Plot vs Descriptive Stats Table



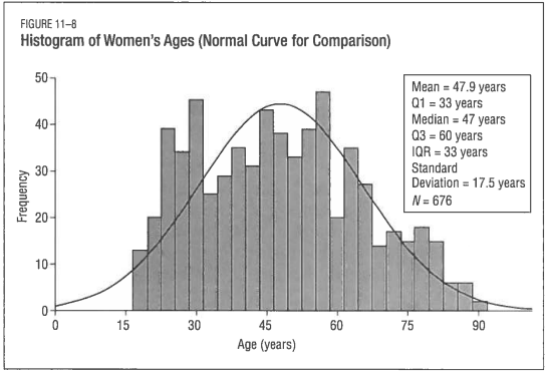

```r
summary(d$CL)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
   1.00   21.00   36.00   35.07   51.00   60.00 
```

# Numerical Variable: Make a Histogram



# Numerical Variable: Make a Histogram

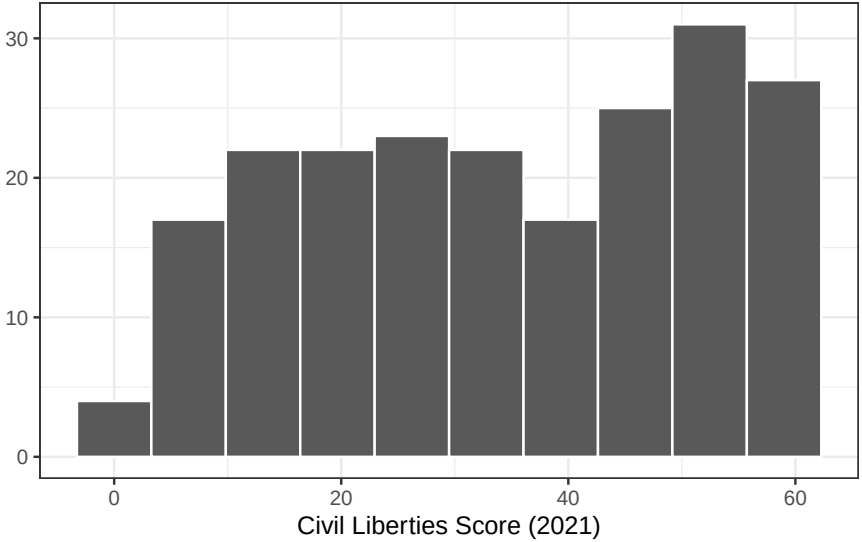



**Why can't we just draw a bar for every possible value? (e.g. make a bar plot)**

# Histograms are bar plots using bins

1. Make a table of `CL` scores for 2021 only
2. Convert this table into bins of 10 and re-sum the counts
   - Bin 1 is all values from 1 - 10
   - Bin 2 is all values from 11 - 20
   - Bin 3...
3. Draw the bar plot of your bins

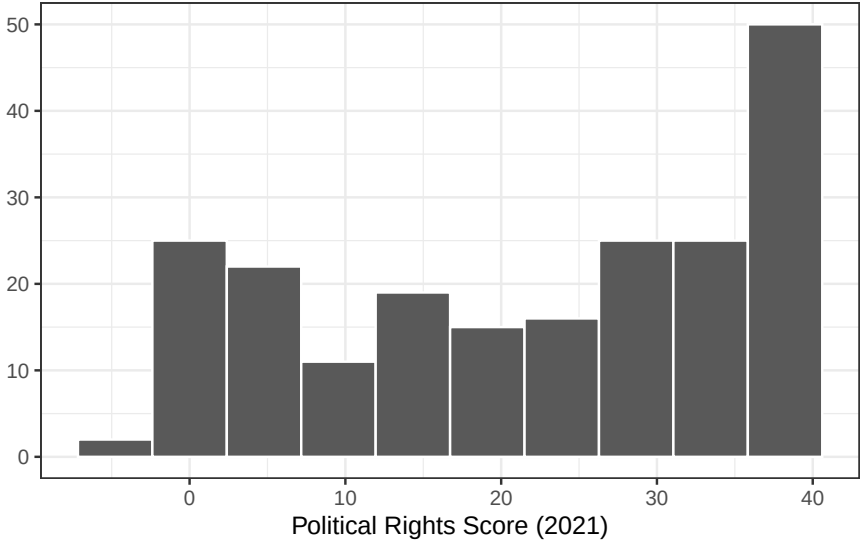
# Numerical Variable: Make a Histogram



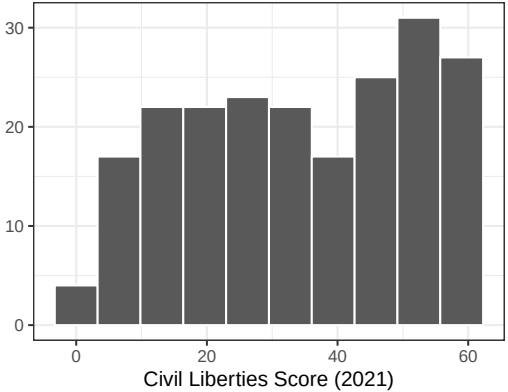
# Histograms are bar plots using bins

1. Make a table of `PR` scores for 2021 only
2. Convert this table into bins of 10 and re-sum the counts
   - Bin 1 is all values from -10 - 0
   - Bin 2 is all values from 1 - 10
   - Bin 3...
3. Draw the bar plot of your bins

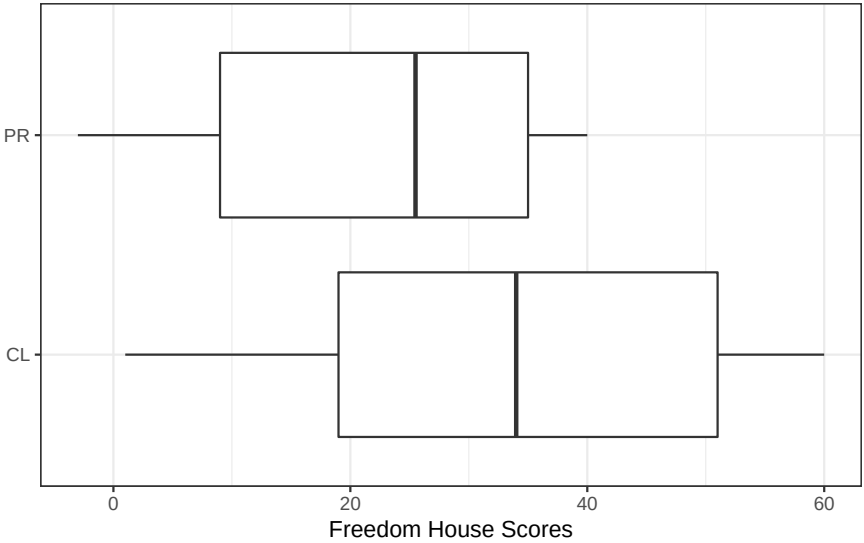
# Numerical Variable: Make a Histogram



# Numeric Variables: Box vs Histogram



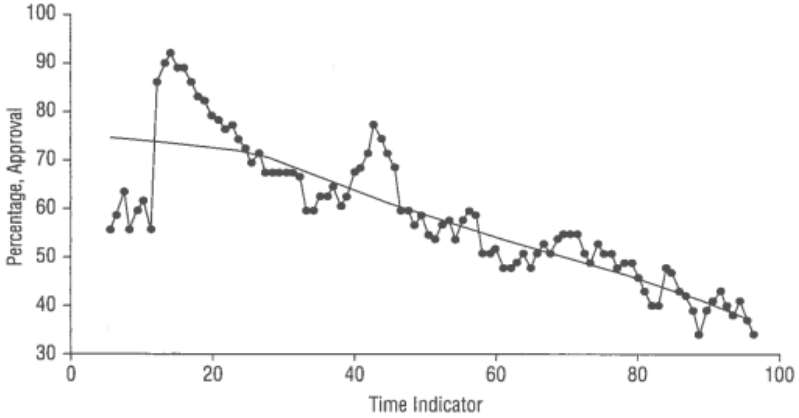
# Box plots make comparing distributions easier



# Numeric Variables Across Time

FIGURE 11–10

Time Series Plot of Presidential Approval Ratings, 2001–2007



# Make a line plot of the average ‘PR’ and ‘CL’ scores per year.

(e.g. two separate lines)

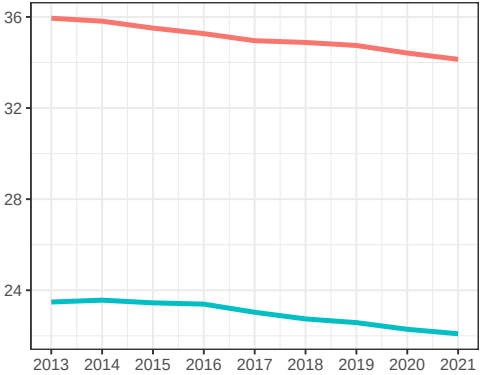
**R Functions**

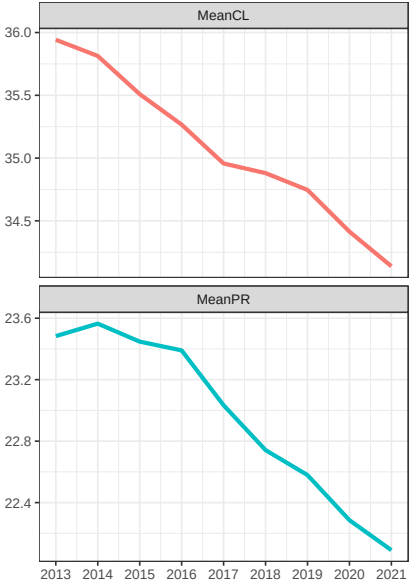
- group_by()
- summarize()
- mean()

# Make a line plot of the average ‘PR’ and ‘CL’ scores per year.

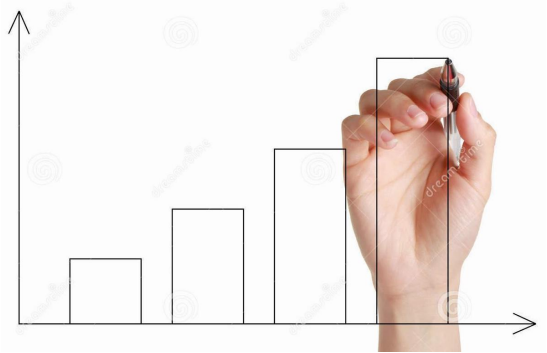
```
d |>
   group_by(Edition) |>
   summarize(
       MeanPR = mean(PR),
       MeanCL = mean(CL)
   )
```

# Which is the better visualization?



# Wrapping Up: Drawing by hand is good!



1. Principle 3 in action,
2. Learn to think like a programmer
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# Principle 3. Variable Type Determines Tool

**Univariate Visualizations**

- If a **categorical** variable, make a **bar plot**
- If a **numeric** variable, make a **box plot** or **histogram**
- If a **numeric variable across time**, make a **line plot**

# For Thursday

Use Healy (3.1 - 3.4) and the Chang recipes to **try** to recreate all of our by-hand plots using R